

数学作业纸

科目_____

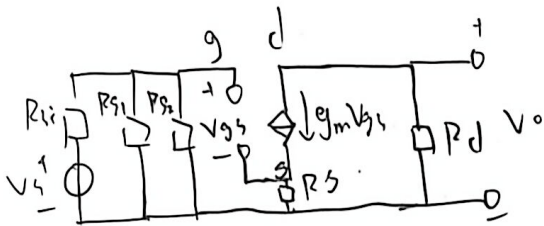
班级:_____

姓名:_____

编号:_____

第 页

$$\begin{aligned}
 4.4.1 \quad g_m &= 2k_n (V_{GS} - V_{TN}) \\
 &= 2 \cdot 1 \text{ mA/V}^2 \times (1.5 - 0.8) \text{ V} \\
 &= 1.4 \text{ mS}
 \end{aligned}$$



$$v_o = -g_m v_{gs} R_D$$

$$v_i = v_{gs} + v_{rs}$$

$$= v_{gs} + g_m v_{gs} R_S$$

$$= v_{gs} (1 + g_m R_S)$$

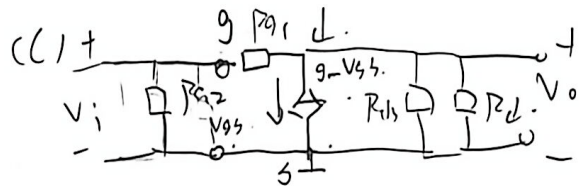
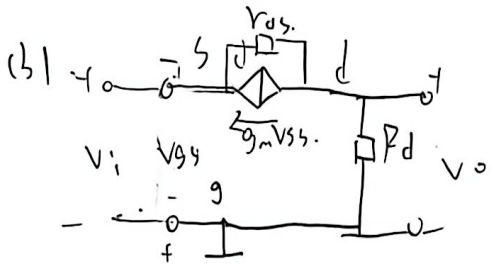
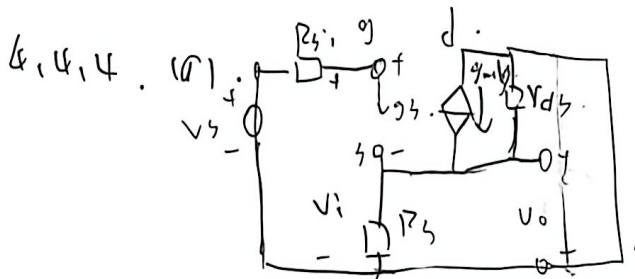
$$A_v = \frac{v_o}{v_i} = \frac{-g_m R_D}{1 + g_m R_S}$$

$$= -8.24$$

$$R_i = R_{G1} \parallel R_{G2} = 8.9 \text{ k}\Omega$$

$$R_o = R_D = 10 \text{ k}\Omega$$

$$A_{vs} = A_v \cdot \frac{R_i}{R_i + R_s} = -8.1$$



$$4.5.3.11. \quad g_m = 2\sqrt{k_p I_{DQ}} \approx 10 \text{ mS}$$

$$R_o = \frac{1}{g_m} = 100 \Omega \quad R_i = R_G = 200 \text{ k}\Omega$$

$$A_v = \frac{g_m R_L}{1 + g_m R_L} \approx 0.91$$

4.8.1 (a) 耗尽型 JFET (b) 耗尽型 NMOS (c) 耗尽型 PMOS (d) 增强型 NMOS

数学作业纸

科目_____

班级: _____

姓名: _____

编号: _____

第 _____ 页

$$4.8.4. A_v = \frac{g_m R_1}{1 + g_m R_1} = 0.92.$$

$$R_1 = R_3 + R_2 \parallel R_4 = 2.075 k\Omega.$$

$$R_0 = R_3 \parallel \frac{1}{g_m} = 1.02 k\Omega.$$